

Instruction:
 1) The ProgModes tab can be copied (as required) to create multiple ProgModes tabs, according to the programming module in the study
 2) Update the "Title" of the ProgModes to be descriptive (e.g. 5A.Baseline Characteristics)
 3) List the newly added ProgModes tab(s) in Section C of the Cover tab
 4) Delete any examples in grey italic text that are not relevant/applicable to the study once the programming specification has been finalized

Title 7.SCT

Section A: Defining Medication Available Period (MAP) medications

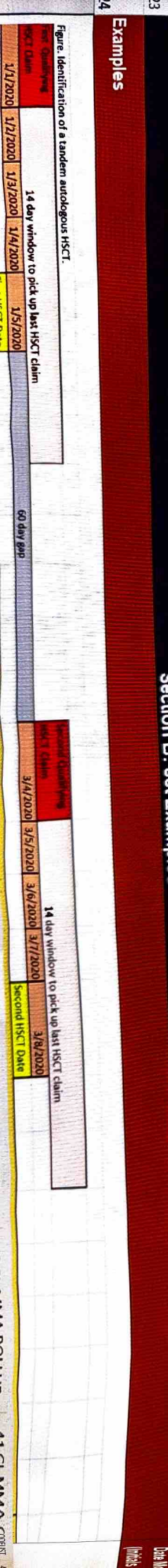
Time Period	Variable	Label	Values	Definition	Code Lists Group	Additional Notes	Date Modified (Initials +)
FU_PD	LOT1_TX_AUTO_DT_1	Date of patient's first transplant after LOT1_START_DT	date, missing	For each patient: Identify a patient's earliest SCT TX_AUTO{t} where: if LOT1_START_DT <= TX_AUTO{t} <= ENDDATE then = TX_AUTO{t}	Tab 47		
FU_PD	LOT1_TX_AUTO_DT_2	Date of patient's second transplant after LOT1_START_DT	date, missing	For each patient: Let the subsequent AUTO SCT date following the AUTO SCT date used to define LOT1_TX_AUTO_DT_1 be TX_AUTO{t+1}. if not missing(LOT1_TX_AUTO_DT_1) then = TX_AUTO{t+1} if missing(TX_AUTO{t+1}) then = missing.	Tab 47		
FU_PD	LOT1_SCT_AUTO_TAND_FLG	Patient had tandem AUTO SCT during LOT1	0,1	if (LOT1_TX_AUTO_DT_2 - LOT1_TX_AUTO_DT_1 + 1) <= 180 and LOT1_TX_AUTO_DT_2 <= ENDDATE and there is not an ALLO SCT such that LOT1_TX_AUTO_DT_1 <= TX_ALLO{t} <= LOT1_TX_AUTO_DT_2 then = 1 Else = 0	Tab 47		
FU_PD	LOT1_SCT_AUTO_SINGLE_FLG	Patient had single AUTO SCT during LOT1	0,1	if not missing(LOT1_TX_AUTO_DT_1) and LOT1_SCT_AUTO_TAND_FLG=0 then = 1 else = 0	Tab 47		

A	B	C	D	E	F	G	H
FU_PD	LOT1_TX_ENDDATE	Potential end date of LOT1 due to SCT-related issue.	date, missing	Let TX_AUTO{+2} be the third AUTO SCT date following the AUTO SCT date used to define LOT1_TX_AUTO_DT_1. if (LOT1_TX_AUTO_DT_2 - LOT1_TX_AUTO_DT_1 + 1) > 180 and LOT1_TX_AUTO_DT_2 <= ENDDATE then = (LOT1_TX_AUTO_DT_2 - 1) else if LOT1_SCT_AUTO_TAND_FLG=1 and not missing(TX_AUTO{+2}) then = TX_AUTO{+2}-1 else do: Iterate over a patient's ALLO SCTs: if missing(LOT1_TX_ENDDATE) or TX_ALLO{f} < LOT1_TX_ENDDATE then = TX_ALLO{f}-1 and stop iterating. Iterate over a patient's CAR-Ts: if missing(LOT1_TX_ENDDATE) or TX_CART{f} < LOT1_TX_ENDDATE then = TX_CART{f}-1 and stop iterating.	Tab 47/48	ALLO SCTs and CAR-T are not allowed to occur during LOT1 and presence of either will immediately end LOT1. In addition, a patient may have a single AUTO or valid tandem SCT. Any auto SCTs occurring after the first auto SCT that are not part of a valid tandem SCT are considered a reason for the LOT to end immediately. Indicates the date LOT1 could potentially end due to an SCT-related issue. For AUTO SCTs: The day before the date of a second transplant if the second transplant occurred more than 180 days after first transplant or the day before the date of the third transplant following a successful tandem transplant. For ALLO SCTs or CAR-T: It is the date of the ALLO SCT or CAR-T occurring prior to any AUTO SCTs that would have been selected earlier to potentially end LOT1.	
15				Let TX_AUTO{+1} be the date of the AUTO SCT following the AUTO SCT date used to define LOT1_TX_AUTO_DT_1. Let TX_AUTO{+2} be the date of the AUTO SCT following TX_AUTO{+1}. if LOT1_TX_ENDDATE = TX_AUTO{+1} or TX_AUTO{+2} then = 1 if LOT1_TX_ENDDATE = TX_ALLO{f}-1 then = 2 if LOT1_TX_ENDDATE = TX_CART{f}-1 then = 3 else = .	Tab 47/48	Indicates a reason for potential LOT1 end: 1 = AUTO SCT reason 2 = ALLO SCT reason 3 = CAR-T reason	
FU_PD	LOT1_TX_ENDDATE _REASON	Reason for LOT1 SCT-related end	1 = AUTO 2 = ALLO 3 = CART . = no TX end reason	Set to the min(LOT1_TX_AUTO_DT_1, LOT1_TX_ENDDATE). If both fields are missing then set to missing. Where: YYYY: AUTO, ALLO or CART A patient may have X AUTO/ALLO transplants where X = 1, 2, 3, ... If a patient has no SCT YYYY claims then their values 5A, MMA_MED 5B, MAP_MED 6, LOT1_BASE 7, SCT 10, LOT1_BASE_END OK 40, CL MMA ROLLUP 41, CL MMA CODELIST 41aMMA_NI	Tab 47		
17	LOT1_1ST_SCT_DT	Date of earliest SCT	date, missing				
FU_PD	LOT1_1ST_SCT_DT	Date of earliest SCT	date, missing				

A	B	C	D	E	F
				Where: YYYY: AUTO, ALLO or CART A patient may have X AUTO/ALLO transplants where $X = 1, 2, 3, \dots$ If a patient has no sct YYYY claims then their values for TX_YYYYX will be set to missing for all X. For each patient: Iterate over the patient's ordered AUTO sct claim's sct_DT_SOURCE: if earliest sct_DT_SOURCE: set TX_AUTO1 to equal max(sct_DT_SOURCE) occurring in (earliest sct_DT_SOURCE + 13) if $X > 1$ and not missing(sct_DT_SOURCE) then: if (sct_DT_SOURCE - TX_AUTO(X-1) = 1) ≥ 60 then: sct_DT_SOURCE is the earliest_date and marks the start of the next 14 day window. Iterate over the claims {} occurring in this window and let TX_AUTOX = sct_DT_SOURCE for claim{} where: earliest_date $\leq$ sct_DT_SOURCE{} $\leq$ earliest_date + 13 and not((earliest_date $\leq$ sct_DT_SOURCE{} + 1) $\leq$ earliest_date + 13) or (sct_DT_SOURCE{} + 1) $\geq$ TX_AUTO(X-1) + 179)) Otherwise set to TX_YYYYX to missing	Starting with multiple sct claims to define, patient level for sct dataset. For each patient: Iterate over the patient's ordered ALLO sct claim's sct_DT_SOURCE and set TX_ALLO1 = sct_DT_SOURCE for the first ALLO claim, TX_ALLO2: sct_DT_SOURCE for the second ALLO claim, ... and so on. Iterate over the patient's ordered CAR-T claim's sct_DT_SOURCE and set TX_ALLO1 = sct_DT_SOURCE for the first CAR-T claim, TX_ALLO2: sct_DT_SOURCE for the second CAR-T claim, ... and so on.
FU_PD	TX_YYYY	Date of YYYY sct X	date, missing		Tab 47/48
18					
19					
20					
21					
22					
23					

Section B: sct Examples

Examples



A

B

C

D

E

F

G

Figure. Identification of a tandem autologous HSCT.

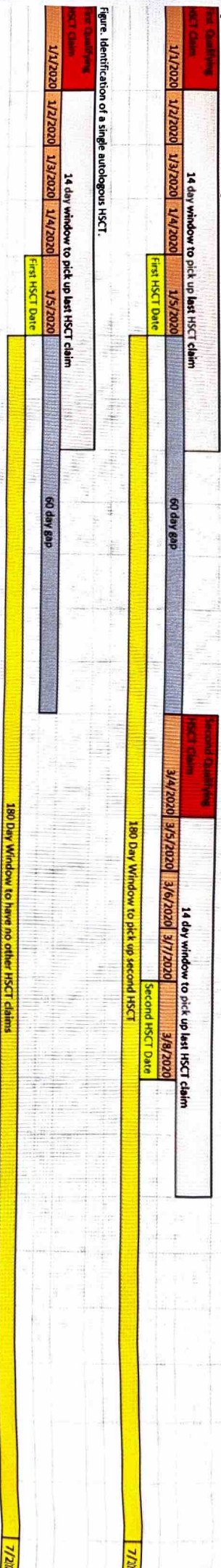


Figure. Identification of a single autologous HSCT.



The patient level sct dataset contains the dates for a patient's processed autologous (auto) and allogeneic (allo) transplant, and CAR-T (cart).

For some patients, there is often a period of sct-related claims activity indicating a workup prior to the actual sct.

For example a patient may have the following sct-related claims (and their dates of service):

Patient A AUTO sct-related dates: 01MAY2018, 02MAY2018, 04MAY2018, 05MAY2018, 07MAY2018, 08MAY2018, 09MAY2018, 06NOV2018, 07NOV2018, 10NOV2018, 11NOV2018, 13NOV2018, 14NOV2018

When processing a patient's sct-related claims and determining the actual sct date, we set the date of the sct to equal the date of last sct claim occurring in a 14 day window from the first sct claim date. In the Patient A example above, 01MAY2018 started the 14 day window during which we would choose the latest sct claim in that 14 day window (the claim on 09MAY2018) as the actual date of Patient A's first sct. Note that while Patient A had additional sct-related claims occurring after 09MAY2018, these claims occurred after the end of the 14 day window.

Furthermore, we require all 'selected' sct dates (the date we've chosen as the actual sct and not workup) to be at least 60 days apart.

In Patient A example above, the first 'selected' sct date is 09MAY2018. Any sct-related claims occurring within 60days of this selected date are not processed (effectively have been skipped over). For Patient A, the next sct-related claim is on 06NOV2018 which is 60+ days after the 'selected' 09MAY2018 sct date and thus will be processed (and in fact starts a new 14 day window).

If the 14 day window overlaps with the date equal to 180 days after the patient's previous sct, then we set the sct to be equal to the claim date that is as close to the 180 day date as possible.

We want to allow for tandem AUTO scts that occur within 180 days of each other. Patient A's first sct date is 09MAY2018, we skip over any sct-related claims within 60days of this 'selected' date (there are not any) and we start a new 14 day window beginning on first sct-related claim on 06NOV2018. We would normally 'select' the last sct-related claim occurring during this 14 day window as the date of the actual sct, but we note that this 14 day window occurs partially within the 180 day tandem window from the selected 09MAY2018 sct date. For simplicity, assume that 09NOV2018 marks 180 days after 09MAY2018. Thus, when processing this group of sct-related claims occurring during the 14 day window beginning on 06NOV2018, we 'select' the claim on 07NOV2018 as the date of the actual sct, even though it is not the latest claim in the 14 day window (14NOV2018). Again, we making this adjustment to allow for a tandem to be appropriately captured (that is, the workup to the tandem started within 180d of our previously selected sct date).

Given that the time window to identify a patient's first sct is between their LOT1_START_DT and ENDDATE, scts identified may end up not being in a patient's LOT1 if their LOT1 ends earlier due to non-sct related reasons.